ECE30030/ITP30010 – Database Systems

Relational Data

Reading: Chapter 1-2

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Agenda

- R-DBMS
- Relational Data Model

Database Systems

- Database
  - Organized collection of inter-related data that models some aspect of the real-world (A. Pavlo)
    - Things related are laid together; *c.f.*, files are not like this

- Database system: *Informal definition*

Magnetic tapes (storage)

* Image src: http://www.gettyimages.com/detail/news-photo/computer-display-with-magnetic-tape-unit-news-photo/3328014#30th-january-1970-ibm-computer-display-with-magnetic-tape-unit-picture-id3328014

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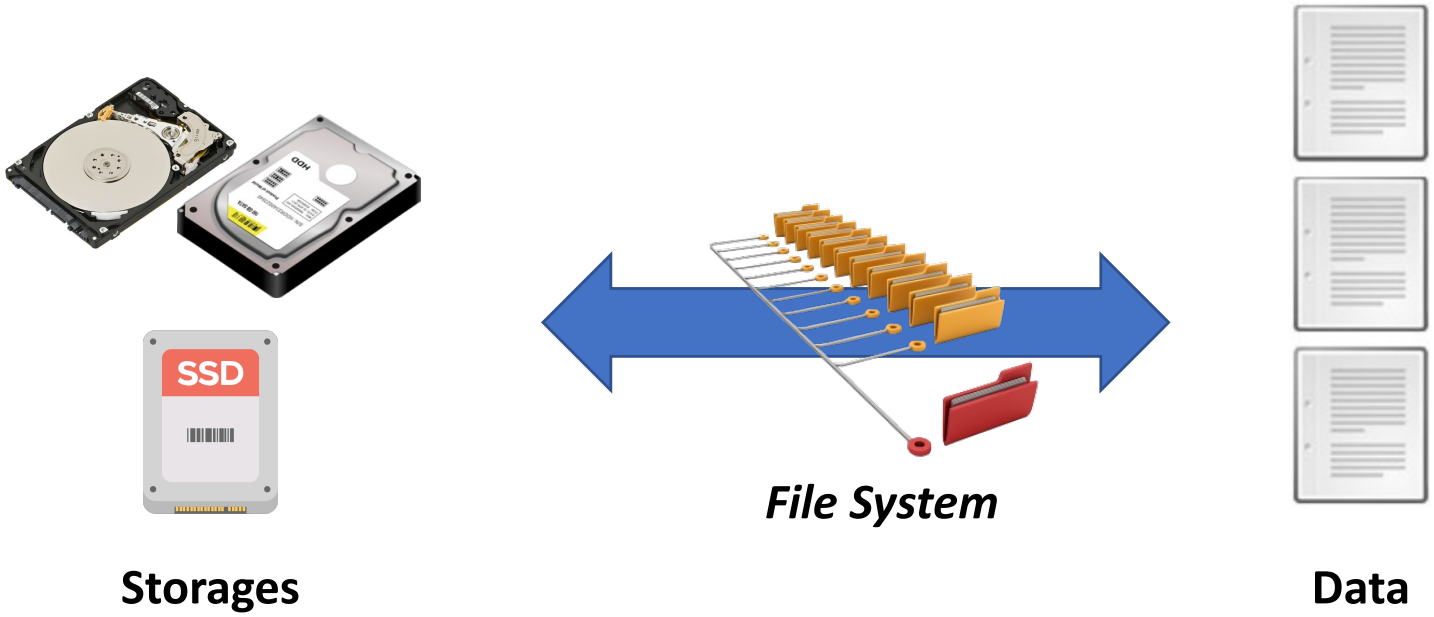

* Image src: https://www.iconfinder.com/icons/4263530/disk_drive_gray_ssd_icon; https://pixabay.com/vectors/hdd-hard-disk-drive-disk-hard-disk-154463/

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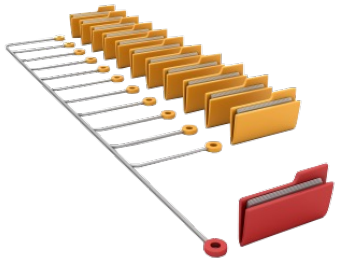

File System

  - Flat file strawman
    - Store a database as comma-separated value (CSV) files
    - Manage the CSV files using our own code
      - Use a separate file per entity
      - The applications have to parse the CSV files each time they want to read or update records

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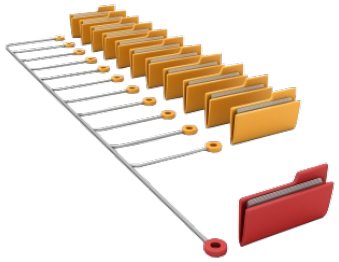

File System

  - Flat file strawman
    - Issue: data integrity
      - How to examine the validity of the values?
    - Issue: implementation
      - How to find a particular record?
      - How to write a new application that uses the same data
    - Issue: durability
      - What if the machine crashes while file writing?

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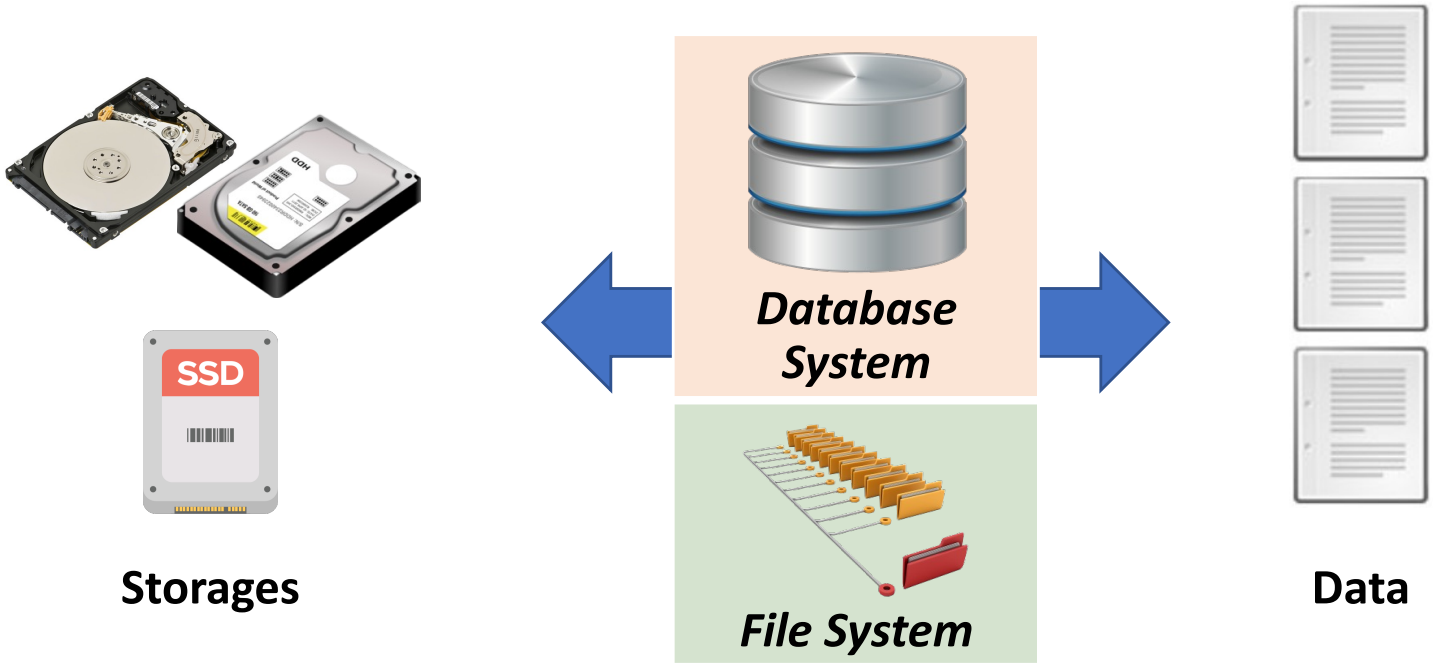

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Database Systems

- Database management system (DBMS)
  - Software that allows applications to store and analyze information in a database
    - Access data without worrying about the file I/O-level details
  - A general-purpose DBMS is designed to allow the definition, creation, querying, update, and administration of databases

Database Systems

- DBMS as a data storage
  - Database abstraction to avoid low-level implementation and maintenance chores
    - Store database in simple data structures
    - Access data through high-level language
  - Database abstraction does not include:
    - How to implement the storage, relations, …
    - Clear separation between logical vs. physical layers

- DBMS as an interface
  - Data definition language (DDL)
  - Data manipulation language (DML)

  → Structured query language (SQL) includes both DDL and DML

Agenda

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Data Model

- Data model: A notion for describing data or information
  - Data model consists of three parts:
    - Structure
    - Operations
    - Constraints
  - Examples
    - Relational data model: the most conventional ← *main focus of the course*!
    - NoSQL
      - Key/value
      - Graph
      - Document
      - Column-family
    - Machine learning
      - Array/matrix
    - Misc.: hierarchical, network

Relational Data Model

- Relational data model: A data model describes data in terms of relations

- Relation
  - An unordered set that contains the relationship of attributes that represent entities

Relation (Table)

- Attribute (column)
  - Attribute values are required to be atomic (indivisible data type)
    - String is an atomic data type in most database systems
  - The set of allowed values for each attribute is called the domain of the attribute
  - NULL is a member of every domain, indicating that the value is "unknown"
    - The NULL values cause complications in many operations

- Tuple (row)
  - A tuple is a set of attribute values (also known as its domain) in the relation
  - Each tuple has one value for each attribute of the relation
  - Values are (normally) atomic/scalar

Example: a *Relation*

- *n*-ary relation = table with *n* columns

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
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4 attributes (columns)

Example: a *Relation*

- *n*-ary relation = table with *n* columns

Header

12 tuples (rows, or records)

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
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15151	Mozart	Music	40000.00
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Relation (Table)

- Relations are unordered: Order of tuples is irrelevant (tuples may be stored in an arbitrary order)
  - Example

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
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76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

ID	name ▲ 2	dept_name ▲ 1	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00
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33456	Gold	Physics	87000.00

Notations

- Using a table

ID	name 2	dept_name 1	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00

- Using a set notation

  Structure: *instructor*(*ID, name, dept_name, salary*),

  Tuples: (*76766, Crick, Biology, 72000.00*),
  (*83821, Brandt, Comp. Sci., 92000.00*),
  (*45565, Katz, Comp. Sci., 75000.00*)

  - Mathematically, sets do not have orders nor duplicates
  - However, we implicitly treat them as ordered sets
    - (*76766, Crick, Biology, 72000.00*) != (*72000.00, Biology, Crick, 76766*)

Keys

- Key
  - One type of constraints
  - One or more attributes form a key
  - A key for a relation → do NOT allow duplicates of the same values of the key attributes

Primary Keys

- A relation's primary key uniquely identifies a single tuple
- Some DBMSs automatically create an internal primary key if you do not define one
  - *E.g.*, SQL:2003 (SEQUENCE), MySQL (AUTO_INCREMENT)

- Example
  - instructor(<u>ID</u>, name, dept_name, salary)

ID	name ▲ 2	dept_name ▲ 1	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00

Foreign Keys

- A foreign key specifies that an attribute from one relation has to map to a tuple in another relation
  - Value in one relation must appear in another relation
    - Referencing relation → Referenced relation

- Example

Relation: instructor

ID	name	dept_name	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00

Relation: department

dept_name	building	budget
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00

Data Language

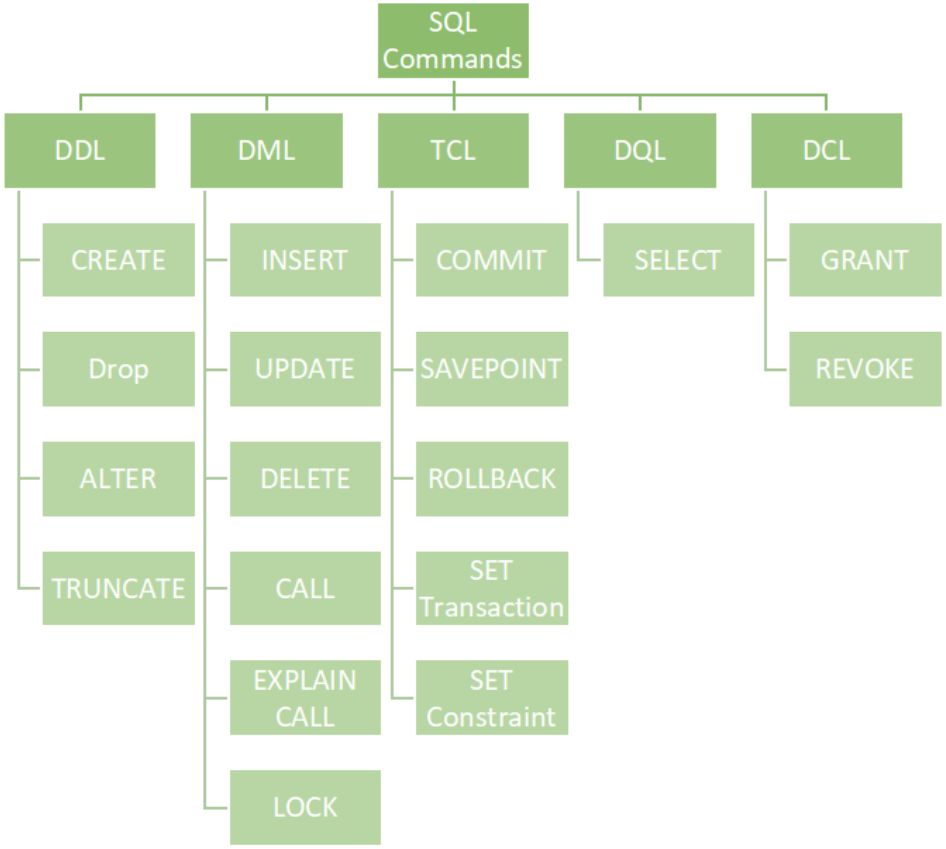
- Data definition language (DDL)
  - How to represent relations and information in a database
    - Defines database schemas

- Data manipulation language (DML)
  - How to store and retrieve information from a database
  - Procedural
    - The query specifies the (high-level) strategy the DBMS should use to find the desired results
    - Based on relational algebra

  - *C.f.*, there are non-procedural DML
    - The query specifies only what data is wanted and not how to find it
    - Based on relational calculus – *this is related to query optimization*

Data Language (Optional)

- A bit more specific …
  - DDL
  - DML
  - TCL: Transaction Control Lang.
  - DQL: Data Query Lang.
  - DCL: Data Control Lang.



Database Schema

- Database: a collection of relations (tables)
- Database schema: the logical structure of the database
- Database instance: a snapshot of the data in the database at a given instant in time
  - Relation instance: a snapshot of a relation (attributes and tuples) at a given instant in time

Next

- Coming next:
  - Relational algebra